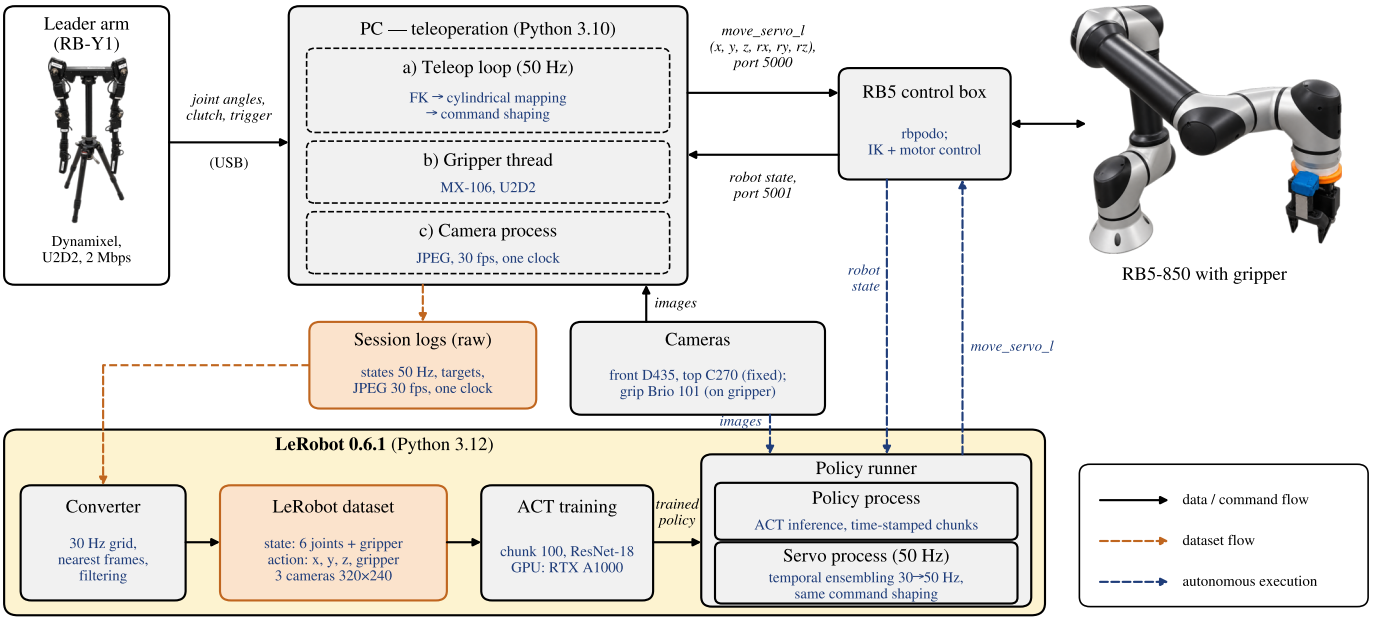

Leader arm
(RB-Y1)
Dynamixel,
U2D2, 2 Mbps
joint angles,
clutch, trigger
(USB)
PC — teleoperation (Python 3.10)
a) Teleop loop (50 Hz)
FK → cylindrical mapping
→ command shaping
b) Gripper thread
MX-106, U2D2
c) Camera process
JPEG, 30 fps, one clock
move_servo_l
(x, y, z, rx, ry, rz),
port 5000
robot state,
port 5001
RB5 control box
rbpodo;
IK + motor control
RB5-850 with gripper
robot
state
move_servo_l
images
Session logs (raw)
states 50 Hz, targets,
JPEG 30 fps, one clock
Cameras
front D435, top C270 (fixed);
grip Brio 101 (on gripper)
images
LeRobot 0.6.1 (Python 3.12)
Converter
30 Hz grid,
nearest frames,
filtering
LeRobot dataset
state: 6 joints + gripper
action: x, y, z, gripper
3 cameras 320×240
ACT training
chunk 100, ResNet-18
GPU: RTX A1000
trained
policy
Policy runner
Policy process
ACT inference, time-stamped chunks
Servo process (50 Hz)
temporal ensembling 30→50 Hz,
same command shaping
data / command flow
dataset flow
autonomous execution